

```
[123]: #PART I
#1.1
my_name = 'Delphin'
my_age = 22
my_gpa = 17.0
am_I_present = True
print(my_name, " : ", type(my_name))
print(my_age, " : ", type(my_age))
print(my_gpa, " : ", type(my_gpa))
print(am_I_present, " : ", type(am_I_present))
```

```
Delphin : <class 'str'>
22 : <class 'int'>
17.0 : <class 'float'>
True : <class 'bool'>
```

```
[78]: #1.2
age_text = "21"
result = (int)(age_text) + 1
print(result)
```

```
22
```

```
[81]: #1.3
print("3"+"4")
#It is not 7, because the operands are strings not integers
```

```
34
```

```
[91]: #PART II
#II.1
f"I am {my_name} {my_age} years old, studying at AUCA"
```

```
[91]: 'I am Delphin 23 years old, studying at AUCA'
```

```
[94]: #II.2.a
course_name = "Big Data Analytics"
course_name = course_name.upper()
print(course_name)
```

```
BIG DATA ANALYTICS
```

```
[93]: #II.2.b
print(len(course_name))
```

```
18
```

```
[105]: #II.2.c
print(course_name[0:3])
```

```
BIG
```

```
[107]: score = [72, 85, 91, 64, 78]
#PART III.1.a
print("The first score is : ",score[0]," ,The last score is : ",score[-1])
```

```
The first score is : 72 ,The last score is : 78
```

```
[109]: #III.1.b
print(len(score))
```

```
5
```

```
[111]: #III.1.c
print(sum(score)/len(score))
```

```
78.0
```

```
[113]: #III.1.d
print("The highest score is ",max(score)," , The lowest score is ",min(score))
```

The highest score is 91 , The lowest score is 64

```
[118]: #PART III.2.a
score = [72, 85, 91, 64, 78]
score.append(88)
print(score)
```

[72, 85, 91, 64, 78, 88]

```
[120]: #III.2.b
score[3]=66
#III.2.c
print(score)
print("New average = ",sum(score)/len(score))
```

[72, 85, 91, 66, 78, 88]
New average = 80.0

```
[125]: #PART IV.1
me = {
    "name":"Delphin",
    "age":22,
    "program":"IT",
    "District":"Gasabo"
}
print(me["program"])
```

IT

```
[126]: #IV.a
me["score"]=18
#IV.b
me["age"] = me["age"]+1
print(me.keys())
```

dict_keys(['name', 'age', 'program', 'District', 'score'])

```
[127]: #IV.c.
print(me)
```

{'name': 'Delphin', 'age': 23, 'program': 'IT', 'District': 'Gasabo', 'score': 18}

```
[99]: import pandas as pd
df = pd.read_excel("week1_students.xlsx")
df.head()

df
```

[99]:

	student_id	name	age	gender	program	district	attendance_pct	score
0	AUCA001	Aline Uwase	21	F	Information Systems	Gasabo	92	85
1	AUCA002	Eric Niyonzima	23	M	Software Engineering	Kicukiro	88	78
2	AUCA003	Diane Mukamana	20	F	Information Systems	Nyarugenge	95	91
3	AUCA004	Jean Bosco Habimana	24	M	Networking	Gasabo	70	64
4	AUCA005	Clarisse Ingabire	22	F	Accounting	Musanze	85	72
5	AUCA006	Patrick Mugisha	21	M	Software Engineering	Huye	90	83
6	AUCA007	Sandrine Umutohi	19	F	Information Systems	Kicukiro	97	94
7	AUCA008	Emmanuel Nsengiyumva	25	M	Networking	Rubavu	65	58
8	AUCA009	Grace Uwimana	20	F	Accounting	Gasabo	82	76
9	AUCA010	Yves Ndayisenga	22	M	Software Engineering	Nyarugenge	78	69
10	AUCA011	Josiane Mukeshimana	21	F	Information Systems	Muhanga	93	88
11	AUCA012	Claude Iradukunda	23	M	Networking	Gasabo	75	66
12	AUCA013	Beata Nyirahabimana	24	F	Accounting	Nyamagabe	89	81
13	AUCA014	Olivier Twagirimana	20	M	Software Engineering	Kicukiro	96	90
14	AUCA015	Chantal Uwera	22	F	Information Systems	Rusizi	68	61
15	AUCA016	Fabrice Byiringiro	21	M	Networking	Musanze	84	74
16	AUCA017	Solange Ishimwe	19	F	Software Engineering	Gasabo	91	87
17	AUCA018	David Hakizimana	25	M	Accounting	Nyagatare	72	55
18	AUCA019	Immaculee Uwamahoro	23	F	Information Systems	Kicukiro	87	79
19	AUCA020	Samuel Rukundo	20	M	Networking	Nyarugenge	94	92

```
[13]: # names = [s["name"] for s in students]
# all_scores = [s["score"] for s in students]
# districts = [s["district"] for s in students]
#a)
total_students = len(df)
print(f"Total number of students is: {total_students}")

Total number of students is: 20
```

```
[21]: #b
round(df['score'].mean(),1)
```

```
[21]: np.float64(77.2)
```

```
[26]: #c)
df['score'].max()
```

```
[26]: 94
```

```
[27]: #c)ii
df['score'].min()
```

```
[27]: 55
```

```
[29]: #d)
range = df['score'].max() - df['score'].min()
print(range)

39
```

```
[42]: #5.2
#a.
df.iloc[0]
```

```
[42]: student_id      AUCA001
name      Aline Uwase
age      21
gender      F
program      Information Systems
district      Gasabo
attendance_pct      92
score      85
Name: 0, dtype: object
```

```
[43]: #b.
df.iloc[-1]
```

```
[43]: student_id      AUCA020
name      Samuel Rukundo
age      20
gender      M
program      Networking
district      Nyarugenge
attendance_pct      94
score      92
Name: 19, dtype: object
```

```
[45]: #c.
name= df.iloc[5]['name']
district= df.iloc[5]['district']
program= df.iloc[5]['program']
score= df.iloc[5]['score']
print(name," from ",district," studies ",program," and scored ", score)

Patrick Mugisha from Huye studies Software Engineering and scored 83
```

```
[66]: Gasabo = df[df["district"] == 'Gasabo'].shape[0]
print("Gasabo : ", Gasabo)
Kicukiro = df[df["district"] == 'Kicukiro'].shape[0]
Nyarugenge = df[df["district"] == 'Nyarugenge'].shape[0]
Rusizi = df[df["district"] == 'Rusizi'].shape[0]
print("Kicukiro : ",Kicukiro)
print("Rusizi : ",Rusizi)

Gasabo : 5
Kicukiro : 4
Rusizi : 1
```

```
[72]: #For top student
score = df['score'].max()
position = df['score'].idxmax()
name = df['name'][position]
print("The top student is ",name," with a score of ",score)

The top student is Sandrine Umutoni with a score of 94
```

```
[ ]: #PART 6
1. IT DOES NOT HAVE VOLUME
```

```
[ ]: 2.I was suprised that we can find the best and the worst performing students without using loops
```

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